

Weather AI Agent

An Intelligent Weather Forecasting Agent built on
IBM watsonx.ai · LangGraph · Meta LLaMA 3

Sagar Kishor Shende

AICTE Intern · Edunet Foundation × IBM | July – August 2025



Internship Overview

Organization

Edunet Foundation
(AICTE × IBM SkillsBuild)

Duration

4 Weeks
14 Jul – 6 Aug 2025

Domain

AI & Cloud
Technologies

Internship Curriculum

Week 1

Orientation, IBM Registration, Cloud Computing, AI Basics

Week 2

Data Analytics, Hands-On EDA, Cloud Data Analytics

Week 3

Chatbot AI, Cloud Chatbot Experiments, ML Problem Statements

Week 4

AI & ML in Cloud, AutoAI on IBM Cloud, Final Project Submission

Internship Offer Letter



Offer Letter Details

Issued To:	Sagar Kishor Shende
Date:	16 July 2025
AICTE Student ID:	STU6844facc380f91749351116
Internship ID:	INTERNSHIP_1748937226683...
Duration:	14 Jul 2025 – 6 Aug 2025
Type:	Virtual / Online
Issued By:	Edunet Foundation (AICTE)
Collaborator:	IBM SkillsBuild



Certification Criteria

- Weekly virtual mentor sessions (mandatory)
- Registration on IBM Academic & IBM Cloud
- Complete min. 2 courses on IBM SkillsBuild
- Submit project presentation



Internship Benefits

- IBM SkillsBuild e-Learning Platform access
- Project-based learning environment
- Masterclasses by Subject Matter Experts
- IBM SkillsBuild & Edunet certification

Problem Statement

The Challenge

- Traditional weather apps give static data — no contextual intelligence.
- Users need a conversational, real-time weather assistant.
- Existing solutions lack natural language understanding and Google Search integration.
- No multi-city, multi-query support in single agent interactions.

60%

of users prefer
conversational AI

3B+

Weather API
requests daily globally

LLM

powered agent for
intelligent responses

Technology Stack

watsonx.ai

IBM's AI platform for
Foundation Model inferencing

Meta LLaMA 3

90B Vision Instruct model
for intelligent responses

LangGraph

ReAct agent framework
with MemorySaver checkpointing

Google Search

Real-time weather data
fetched via search integration

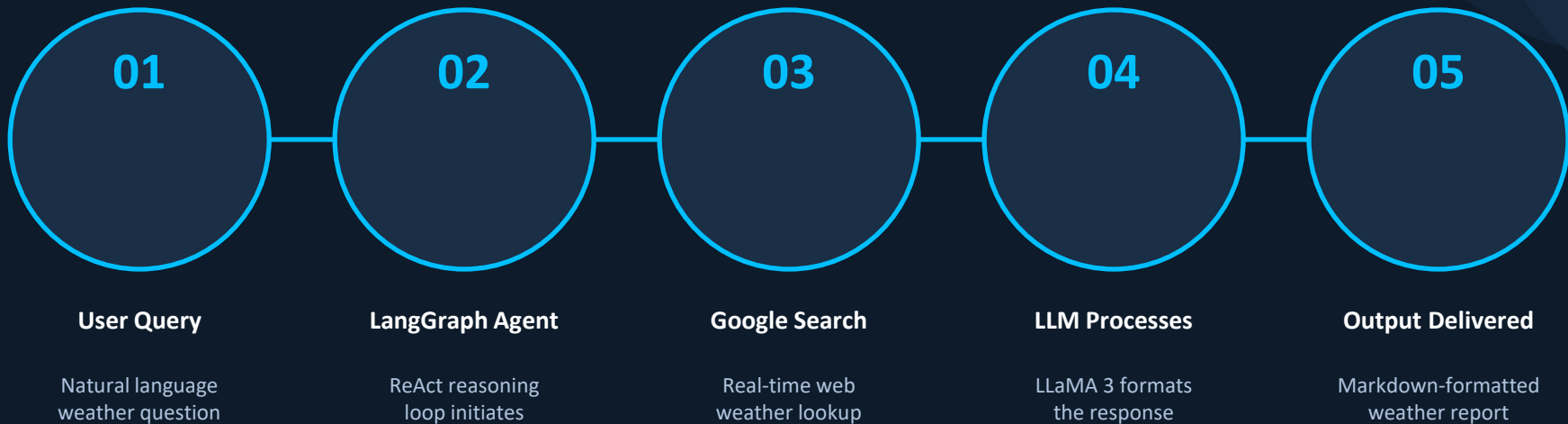
IBM Cloud

Cloud deployment &
API key management

Python + LangChain

langchain_ibm, ibm_watsonx_ai
for orchestration

Agent Architecture & Workflow



🧠 **MemorySaver:** Conversation state persisted across turns via LangGraph checkpointing — agent remembers context for multi-turn interactions.

Key Features



Multi-City Support

Ask weather for any city in India or globally — Mumbai, Tokyo, New York and more.



Detailed Reports

Temperature, humidity, wind, rain probability, sunrise/sunset, and expert advice.



LLM-Powered Intelligence

LLaMA 3 (90B) understands natural language queries and formats contextual answers.



Real-Time Google Search

Live weather data fetched on every query — no stale cached data.



Multi-Turn Memory

MemorySaver checkpointing lets agent remember past exchanges in same session.



Markdown Output

Clean, formatted reports with emojis, sections, and actionable suggestions.

Sample Agent Output

User: What is the weather in Nagpur today?

Agent Response



Weather Report — Nagpur, Maharashtra (7th August 2025)



Current Weather:

Temperature: 31°C | Condition: Partly Cloudy | Feels Like: 34°C | Humidity: 66%



Rain Update: Light rain expected in the evening (6 PM – 8 PM) | Rain Probability: 60%



Today's Range: High 33°C / Low 25°C



Sunrise: 5:54 AM



Sunset: 6:48 PM



Expert Suggestion: Carry an umbrella. Stay hydrated — humidity is high.

Learnings & Outcomes

Technical Learnings

- Built and configured a LangGraph ReAct agent with IBM watsonx.ai
- Integrated Google Search as a live tool for real-time data
- Learned IBM Cloud API key management and project ID configuration
- Implemented MemorySaver for stateful multi-turn conversations
- Worked with Foundation Model parameters (temperature, top_p, max_tokens)
- Structured agent instructions for markdown-formatted outputs

Certifications & Achievements



Certificate of Completion

Edunet Foundation × AICTE × IBM



IBM SkillsBuild Badge

Getting Started with AI — Jul 22, 2025



Project Submitted

Weather AI Agent using IBM watsonx.ai



4-Week Internship

Jul 15 – Aug 7, 2025 | AI & Cloud

IBM SkillsBuild — Certificate of Completion

IBM
SkillsBuild

Certificate of Completion

This is to certify that

SAGAR SHENDE

has successfully completed 4-week Internship, leveraging SkillsBuild & IBM Cloud Platform in Emerging Technologies (AI & Cloud) from 15th July 2025 to 7th August 2025

IBM SkillsBuild — Getting Started with AI Badge

Getting Started with Artificial Intelligence

IBM SkillsBuild



In recognition of the commitment to achieve professional excellence

Sagar Shende

Has successfully satisfied the requirements for:

Getting Started with Artificial Intelligence

Issued on: Jul 22, 2025 | Issued by: IBM SkillsBuild

Verify: credly.com/badges/570287dd-80ff-4337-a7e2-7d695b21ff44

IBM SkillsBuild — RAG with LangChain Certificate

IBM SkillsBuild

Completion Certificate |
SkillsBuild

Completion Certificate

This certificate is presented to

Sagar Shende

for the completion of

Lab: Retrieval Augmented Generation with LangChain

(ALM-COURSE_3824998)



Thank You!

Built with IBM watsonx.ai · LangGraph · Meta LLaMA 3 · Google Search

Sagar Kishor Shende

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IBM SkillsBuild

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Edunet Foundation

AI & Cloud 2025